

Time Series Foundations - Bihag

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Introduction

Part 0: Setting up RMarkdown in RStudio

Step 1: Create RMarkdown File

1. Click File → New File → R Markdown
2. Title: Time Series Foundations
3. Output: HTML (recommended)
4. Click OK

Step 2: Installing the required packages

Note: you only need to do this once.

```
install.packages("forecast") install.packages("tseries") install.packages("ggplot2") install.packages("urca")
install.packages("TSA")
```

Step 3: Understanding RMarkdown Chunks.

RMarkdowns allows combining of text+code+output

```
# Your R code here. (The hashtag means it is a comment)
```

Keyboard Shortcut:

Mac: Command + Option + I

Windows: Ctrl + Alt + I

Part 1: Loading Libraries

Create a chunk named setup and load the libraries to be used.

```
library(forecast)
```

```
## Warning: package 'forecast' was built under R version 4.4.3
```

```
## Registered S3 method overwritten by 'quantmod':
```

```
##   method             from
```

```
##   as.zoo.data.frame zoo
```

```
library(tseries)
```

```
## Warning: package 'tseries' was built under R version 4.4.3
```

```
library(ggplot2)

## Warning: package 'ggplot2' was built under R version 4.4.3
library(urca)

## Warning: package 'urca' was built under R version 4.4.3
library(TSA)

## Warning: package 'TSA' was built under R version 4.4.3
## Registered S3 methods overwritten by 'TSA':
##   method      from
##   fitted.Arima forecast
##   plot.Arima   forecast
##
## Attaching package: 'TSA'
## The following objects are masked from 'package:stats':
##
##   acf, arima
## The following object is masked from 'package:utils':
##
##   tar
```

Part 2: Overview of Learning Goals

In this practical session, the students will use available dataset in R to:

1. Create Time Series Graphics
2. Perform Statistical Tests for Trend, Drift, Seasonality
3. Perform ADF Stationarity Test
4. Interpret results

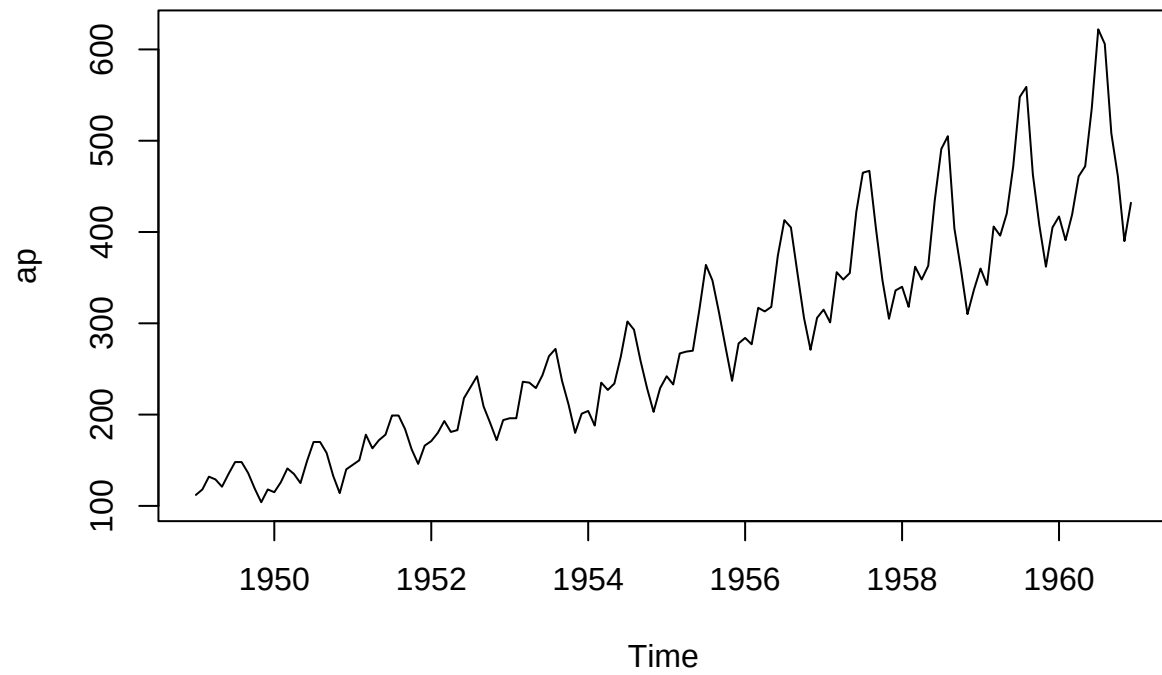
Example 1: Air Passengers Dataset

Data Info: Monthly Airline Passenger Numbers 1949-1960. The classic Box & Jenkins airline data. Monthly totals of international airline passengers, 1949 to 1960.

```
# Step 1: Load dataset
data("AirPassengers")
ap <- AirPassengers

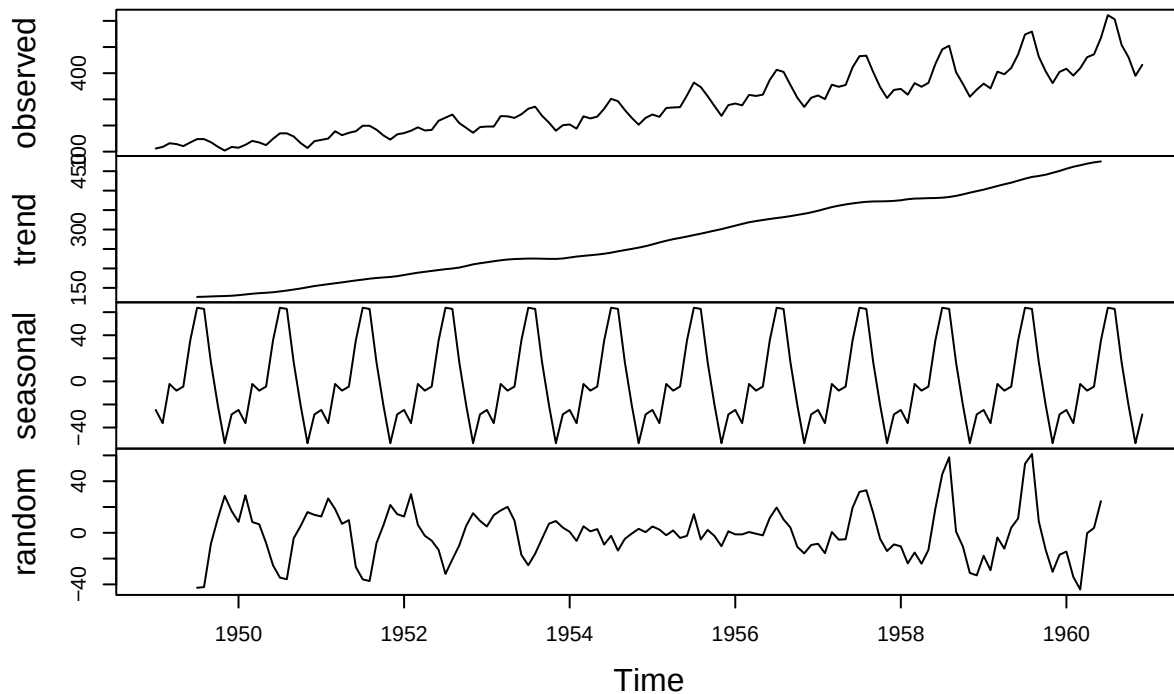
# Step 2: Time series plot
plot(ap, main="AirPassengers Time Series")
```

AirPassengers Time Series



```
# Step 3: Decomposition (Trend + seasonality Test)  
decomp_ap <- decompose(ap)  
plot(decomp_ap)
```

Decomposition of additive time series



```
# Step 4: Drift Test (Mean Change Over Time)
```

```
t.test(window(ap, end = c(1955,12)),
window(ap, start = c(1956,1)))
```

```
##
##  Welch Two Sample t-test
##
## data:  window(ap, end = c(1955, 12)) and window(ap, start = c(1956, 1))
## t = -16.273, df = 101.7, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -223.3561 -174.8201
## sample estimates:
## mean of x mean of y
##  197.3452  396.4333
```

```
# Step 5: ADF Test
```

```
adf.test(ap)
```

```
## Warning in adf.test(ap): p-value smaller than printed p-value
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  ap
## Dickey-Fuller = -7.3186, Lag order = 5, p-value = 0.01
## alternative hypothesis: stationary
```

```
# Step 6: Data Transformations
```

```
# Step 6.1: Log transform the data and test
```

```
log_ap <- log(ap)
```

```
adf.test(log_ap)
```

```
## Warning in adf.test(log_ap): p-value smaller than printed p-value
```

```
##
```

```
## Augmented Dickey-Fuller Test
```

```
##
```

```
## data: log_ap
```

```
## Dickey-Fuller = -6.4215, Lag order = 5, p-value = 0.01
```

```
## alternative hypothesis: stationary
```

```
# Step 6.2: Difference the data and test
```

```
d1_ap <- diff(ap)
```

```
adf.test(d1_ap)
```

```
## Warning in adf.test(d1_ap): p-value smaller than printed p-value
```

```
##
```

```
## Augmented Dickey-Fuller Test
```

```
##
```

```
## data: d1_ap
```

```
## Dickey-Fuller = -7.0177, Lag order = 5, p-value = 0.01
```

```
## alternative hypothesis: stationary
```

Example 2: LakeHuron Dataset

Data Info: Level of Lake Huron 1875–1972. Annual measurements of the level, in feet, of Lake Huron 1875–1972.

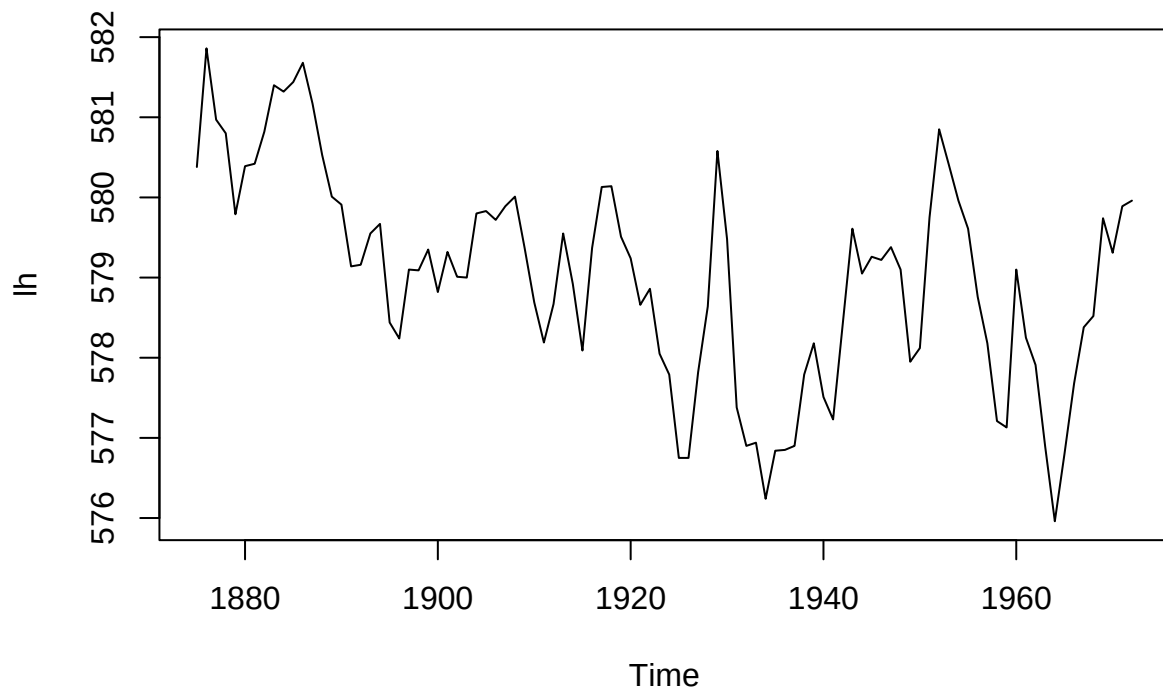
```
# Step 1: Load dataset and plot the series
```

```
data("LakeHuron")
```

```
lh <- LakeHuron
```

```
plot(lh, main="Lake Huron Water Levels")
```

Lake Huron Water Levels



```
# Step 2. ADF test.
```

```
adf.test(lh)
```

```
##
```

```
## Augmented Dickey-Fuller Test
```

```
##
```

```
## data: lh
```

```
## Dickey-Fuller = -2.7796, Lag order = 4, p-value = 0.254
```

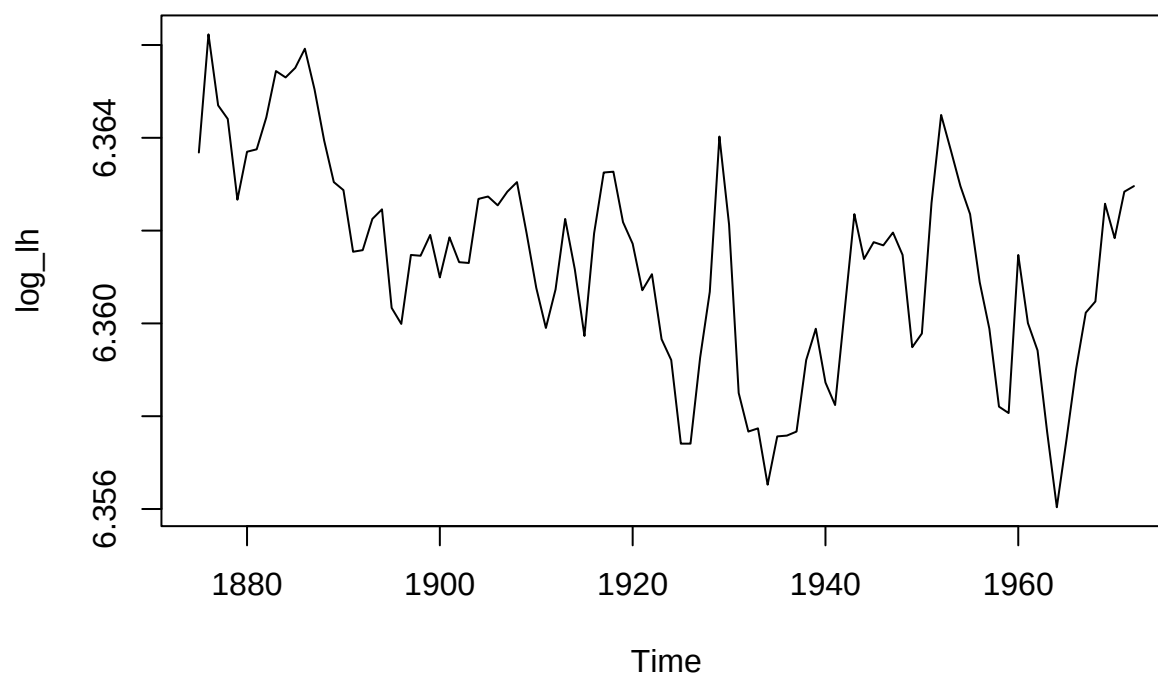
```
## alternative hypothesis: stationary
```

```
# Step 3: Data Transformation (Log)
```

```
log_lh <- log(lh)
```

```
plot(log_lh, main="Log-transformed Lake Huron Water Levels")
```

Log-transformed Lake Huron Water Levels



```
adf.test(log_lh)
```

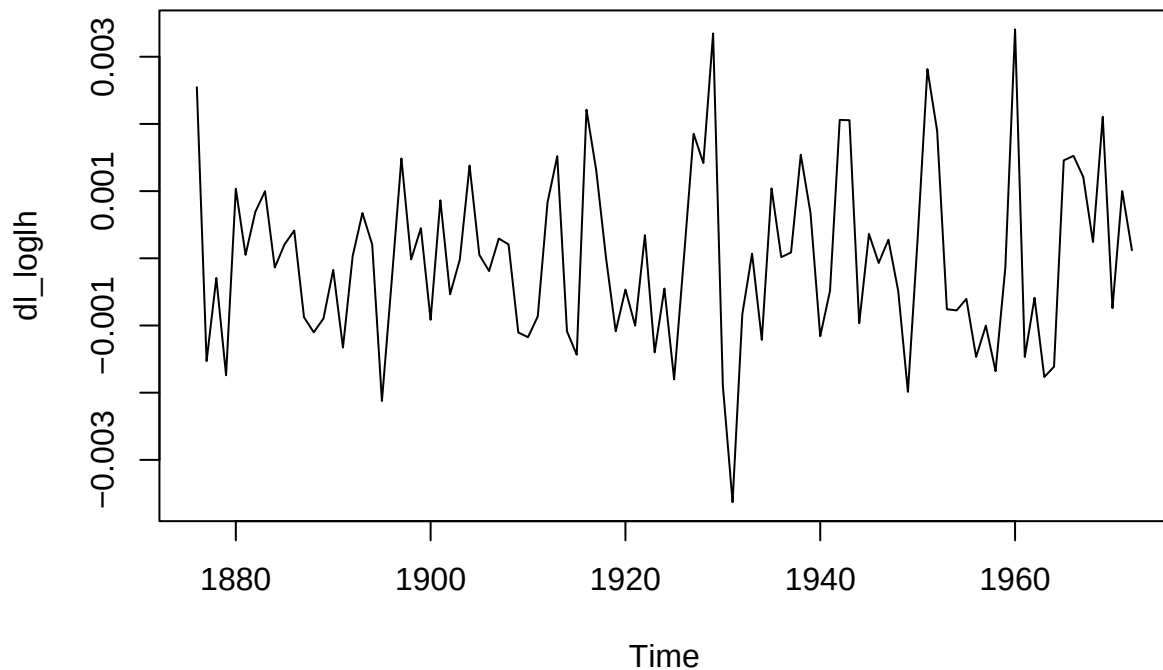
```
##  
## Augmented Dickey-Fuller Test  
##  
## data: log_lh  
## Dickey-Fuller = -2.7792, Lag order = 4, p-value = 0.2542  
## alternative hypothesis: stationary
```

```
# Step 4: Data transformation (difference)
```

```
dl_loglh <- diff(log_lh)
```

```
plot(dl_loglh, main="1st Difference of Log-transformed Lake Huron Water Levels")
```

1st Difference of Log-transformed Lake Huron Water Levels



```
adf.test(dl_loglh)
```

```
## Warning in adf.test(dl_loglh): p-value smaller than printed p-value
```

```
##
```

```
## Augmented Dickey-Fuller Test
```

```
##
```

```
## data: dl_loglh
```

```
## Dickey-Fuller = -5.4689, Lag order = 4, p-value = 0.01
```

```
## alternative hypothesis: stationary
```

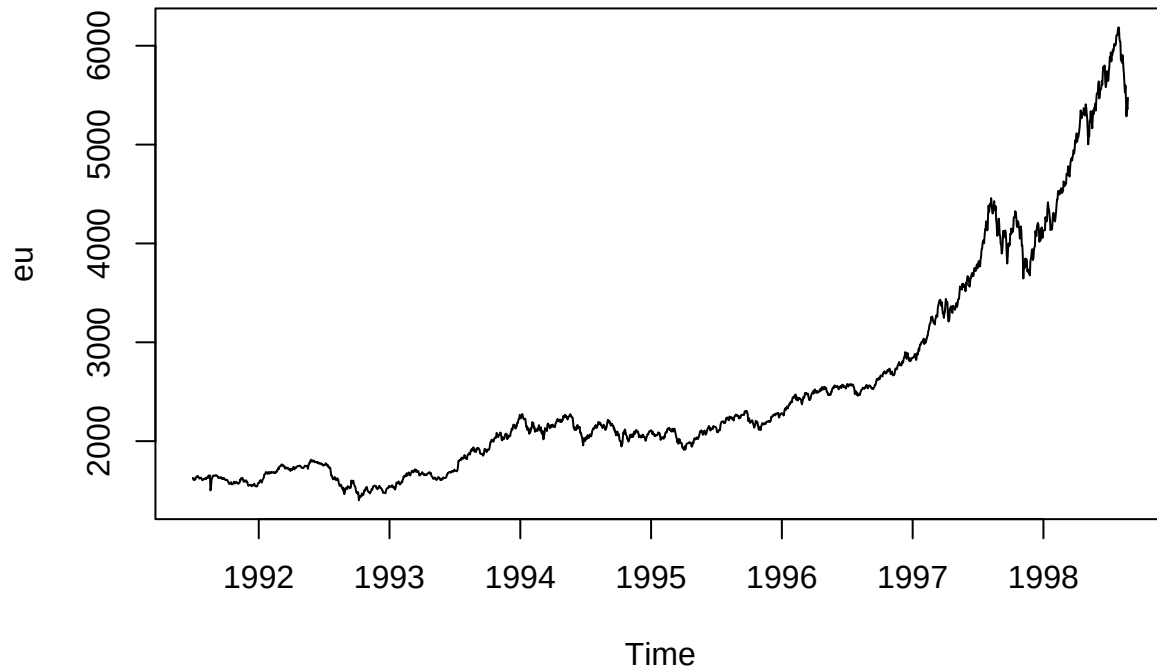
Practical Session 1

Data Info: Daily Closing Prices of Major European Stock Indices, 1991–1998 Contains the daily closing prices of major European stock indices: Germany DAX (Ibis), Switzerland SMI, France CAC, and UK FTSE. The data are sampled in business time, i.e., weekends and holidays are omitted.

Loading dataset

```
data("EuStockMarkets")
eu <- EuStockMarkets[, "DAX"]
plot(eu, main="EU Stock Markets (DAX)")
```


EU Stock Markets (DAX)



Stationarity Testing

We can test if the dataset is stationary or not using the Augmented Dickey-Fuller test. This tests the data against the null hypothesis of it being nonstationary.

```
adf.test(eu)

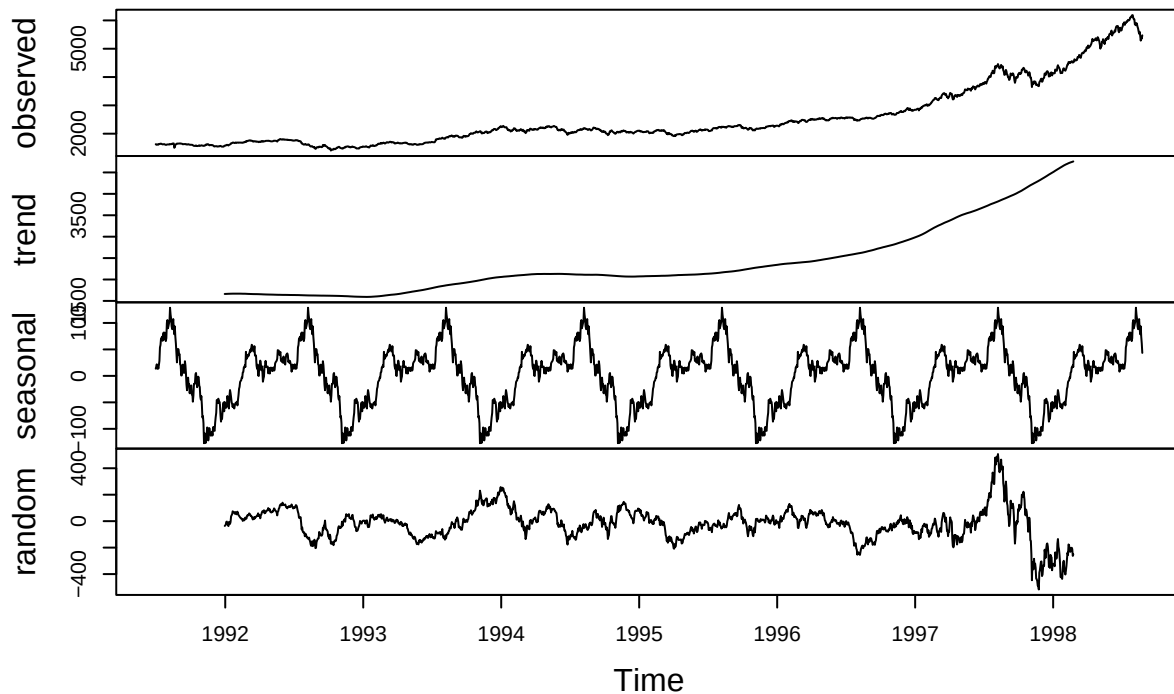
##
## Augmented Dickey-Fuller Test
##
## data: eu
## Dickey-Fuller = -0.82073, Lag order = 12, p-value = 0.9598
## alternative hypothesis: stationary
```

With the p-value being more than 0.05, we fail to reject the null hypothesis—the data is found to be stationary.

Time Series Decomposition

```
decomp_eu <- decompose(eu)
plot(decomp_eu)
```

Decomposition of additive time series



From the Time Series Decomposition graph, there seems to be an upward trend going on, with a steepness increasing from around 1997 to 1998. Seasonality can also be observed, wherein it dips around the end of the year and peaks at around 2/3.

The ADF test performed found the data to be non-stationary. Because of this, data transformations must be made to continue our analysis.

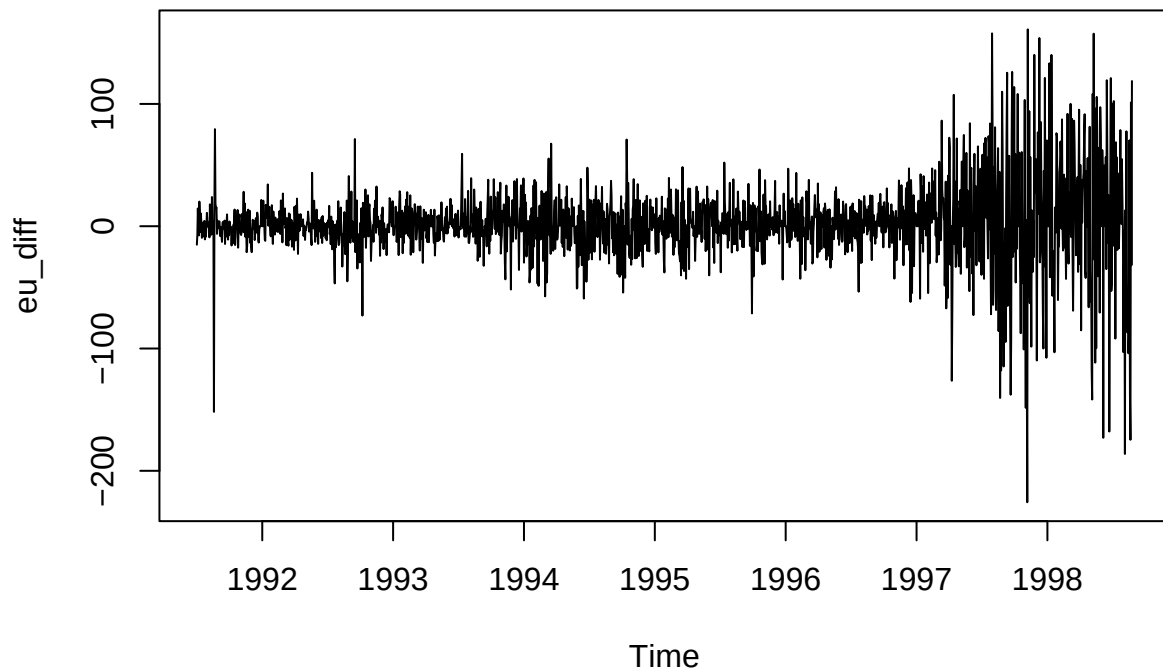
Data Transformation

Transformations to the data can help it be stationary. The two common transformation methods are Log Transformation and Differencing. The former is great in helping stabilizing variance. This method is for data whose variance appear to be changing over time (data becomes “wider”). The latter is great for removing the trend and seasonality of the data. This is great for data that exhibit both.

For our data (EU Stock Market-DAX), there appears to be no changes in variance over time, only trend and seasonality. Because of this, differencing is deemed to be the appropriate method to making the data stationary.

```
eu_diff <- diff(eu)
plot(eu_diff, main="1st Difference of EU Stock Market-DAX")
```

1st Difference of EU Stock Market-DAX



```
adf.test(eu_diff)
```

```
## Warning in adf.test(eu_diff): p-value smaller than printed p-value
```

```
##
```

```
## Augmented Dickey-Fuller Test
```

```
##
```

```
## data: eu_diff
```

```
## Dickey-Fuller = -9.9997, Lag order = 12, p-value = 0.01
```

```
## alternative hypothesis: stationary
```

After the transformation, the ADF test resulted a $p\text{-value} < 0.05$, thus we reject the null hypothesis. The 1st differenced data is stationary.